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REVIEW

Artificial intelligence in medicine: a scoping review of the risk of deskilling and loss of expertise among physicians

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Available online xxx

Background: Artificial intelligence (AI) systems are increasingly deployed in clinical practice, particularly in radiology, pathology, endoscopy, and decision support. While these tools improve efficiency and accuracy, concerns have arisen about deskilling—the erosion of physicians' expertise due to reliance on automation.

Materials and methods: We conducted a narrative review of empirical studies, randomized trials, and theoretical analyses published up to August 2025. The focus was on quantitative evidence of decreased performance following AI exposure, automation bias, and structural changes in training environments. Sources included PubMed, Embase, and gray literature.

Results: Evidence of clinical deskilling, though scarce, is consistent across specialties. In a multicenter randomized trial in colonoscopy, the adenoma detection rate (ADR) dropped significantly from 28.4% to 22.4% when endoscopists reverted to non-AI procedures after repeated AI use, while ADR remained stable with AI assistance (25.3%). In radiology, a controlled study of 27 breast imaging radiologists showed that erroneous AI prompts increased false-positive recalls by up to 12%, even among experienced readers. In computational pathology, experimental web-based tasks revealed that over 30% of participants reversed correct initial diagnoses when exposed to incorrect AI suggestions under time constraints. Structural deskilling has been reported in cytology following the UK's transition to human papillomavirus primary screening, leading to an 80%-85% reduction in case volumes and consolidation of laboratories from 45 to 8 centers, with major implications for training capacity. Across domains, analyses confirm the presence of automation bias and highlight risks of diminished independent diagnostic reasoning.

Conclusions: Although limited in number, empirical studies consistently demonstrate that AI can inadvertently impair physicians' performance or reduce opportunities for skill maintenance. Quantitative evidence of decreased diagnostic accuracy, error propagation, and training erosion underscores the need for longitudinal monitoring, adaptive curricula, and regulatory frameworks to mitigate deskilling. Safeguarding clinical expertise should be considered a central component of AI safety and resilience in medicine.

Key words: artificial intelligence, clinical deskilling, automation bias, medical expertise erosion

INTRODUCTION

The integration of artificial intelligence (AI) into clinical practice is rapidly transforming the landscape of modern medicine. Algorithms capable of interpreting medical images, predicting treatment outcomes, or supporting therapeutic decision making are increasingly being deployed across diverse specialties, from radiology and pathology to dermatology and oncology.[1] Proponents argue that these technologies have the potential to enhance diagnostic accuracy, increase efficiency, and reduce variability in care. However, alongside these anticipated benefits, concerns have emerged regarding unintended consequences for the development and maintenance of physicians' expertise.[2]

A central concern is the risk of reduced independent performance when AI support is withdrawn or unavailable. This phenomenon is often discussed as 'deskilling', i.e. the erosion of competencies through decreased practice, cognitive offloading, and a shift from primary task execution to supervisory monitoring.[3] Importantly, emerging evidence also suggests a broader recomposition of skills ('skill shift'): clinicians may gain new competencies (e.g. tool calibration, error detection, workflow orchestration), while foundational perceptual, interpretive, or procedural skills may receive less reinforcement. Similar dynamics

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have been described in other high-reliability domains where automation changes the balance between hands-on practice and oversight.[4,5]

Recent literature has expanded the understanding of deskilling by introducing a typology of its manifestations: technical (loss of manual skills), cognitive (reduced reasoning), moral (diminished ethical judgment), social (weakened patient communication), and semiotic (reduced sensitivity to clinical signs).[3] These dimensions reflect how AI may not only affect individual competencies but also reshape team dynamics, organizational routines, and professional identity. Several authors have already warned about these dynamics. Cabitza et al.[6,7] emphasized the unintended consequences of automation in medicine, including automation bias and cognitive dependence. Greenhalgh et al.[8,9] demonstrated, through the Non-Adoption, Abandonment, Scale-Up, Spread and Sustainability framework, that the adoption of technologies often entails profound reconfigurations of professional roles, sometimes at the expense of clinical competence. Mittelstadt and Floridi[10] raised ethical concerns regarding the overreliance on data-driven systems, while Topol[11] highlighted the paradox that AI may 'give time back to doctors' yet simultaneously threaten the core identity of medical expertise.

In clinical practice, deskilling can take several forms. Trainees may experience reduced experiential learning when algorithms provide immediate answers, limiting their opportunities to consolidate diagnostic and procedural expertise.[12] More experienced physicians may become cognitively dependent on automated recommendations, gradually weakening their independent judgment and adaptability—especially when facing rare or atypical cases outside the scope of training datasets. This phenomenon involves both cognitive offloading and a disruption of the feedback loops essential for skill acquisition and maintenance. Over time, such dynamics may erode not only technical proficiency but also the humanistic and ethical dimensions of medical expertise.[13,14]

The objective of this scoping review is therefore to synthesize existing knowledge on the risk of deskilling and the potential loss of expertise among physicians in the era of AI. By mapping the current literature, identifying affected specialties, and highlighting gaps, this review aims to provide a foundation for future research and inform strategies to safeguard and adapt medical expertise in an increasingly automated clinical environment.

MATERIALS AND METHODS

This scoping review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) guidelines. The protocol was developed a priori to ensure methodological transparency and reproducibility.

We considered for inclusion all publications addressing the risk of deskilling, loss of expertise, automation bias, or erosion of diagnostic, interpretative, or procedural skills among physicians in the context of AI adoption. Eligible studies were restricted to those published in the last 10 years, between January 2015 and March 2025, as this period corresponds to the emergence and widespread diffusion of clinically relevant AI applications in medicine. Only articles written in English were considered. Eligible study designs included original empirical research, whether quantitative, qualitative, or mixed methods, as well as systematic or narrative reviews, editorials, and conceptual papers. Gray literature, including conference proceedings and reports, was also screened when relevant. Studies focusing exclusively on the technical development of algorithms without reference to physician expertise, or those concerning non-medical populations, were excluded.

A comprehensive search strategy was developed and applied across multiple electronic databases, including PubMed/Medline, Embase, Web of Science, and Scopus. Gray literature was identified using Google Scholar and the proceedings of major medical conferences. The search combined controlled vocabulary terms [Medical Subject Headings (MeSH) and Emtree] with free-text key words related to AI, machine learning, deep learning, clinical decision support, deskilling, automation bias, medical expertise, and physician skills. An example of the PubMed query is as follows: ("artificial intelligence"[MeSH Terms] OR "machine learning"[MeSH Terms] OR "deep learning"[Title/Abstract] OR "clinical decision support"[Title/Abstract]) AND ("deskilling"[Title/Abstract] OR "loss of competence"[Title/Abstract] OR "automation bias"[Title/Abstract] OR "medical expertise"[Title/Abstract] OR "skill erosion"[Title/Abstract]) AND ("physicians"[MeSH Terms] OR "medical doctors"[Title/Abstract] OR "trainees"[Title/Abstract]) AND ("2015/01/01"[Date - Publication] : "2025/03/31"[Date - Publication]). The PRISMA flowchart is presented in Figure 1.

Data were extracted using a standardized form including bibliographic details, medical specialty, AI application type, affected skills, study design, and mitigation strategies. The synthesis was descriptive and thematic, supported by Table 1 and Figure 1. No formal risk-of-bias assessment was carried out, in line with scoping review methodology.

RESULTS

The systematic search identified a growing body of literature published between 2015 and 2025 addressing the potential loss of physician expertise in relation to AI adoption. Evidence included empirical studies in specific specialties, structural changes in practice exposing risks of skill erosion, and conceptual or ethical analyses of automation bias and deskilling. Overall, the number of high-quality empirical studies remains limited, but consistent themes emerge across domains.

Gastroenterology

The strongest empirical signal of clinical deskilling was observed in colonoscopy. A multicenter observational study conducted in Poland and published in The Lancet

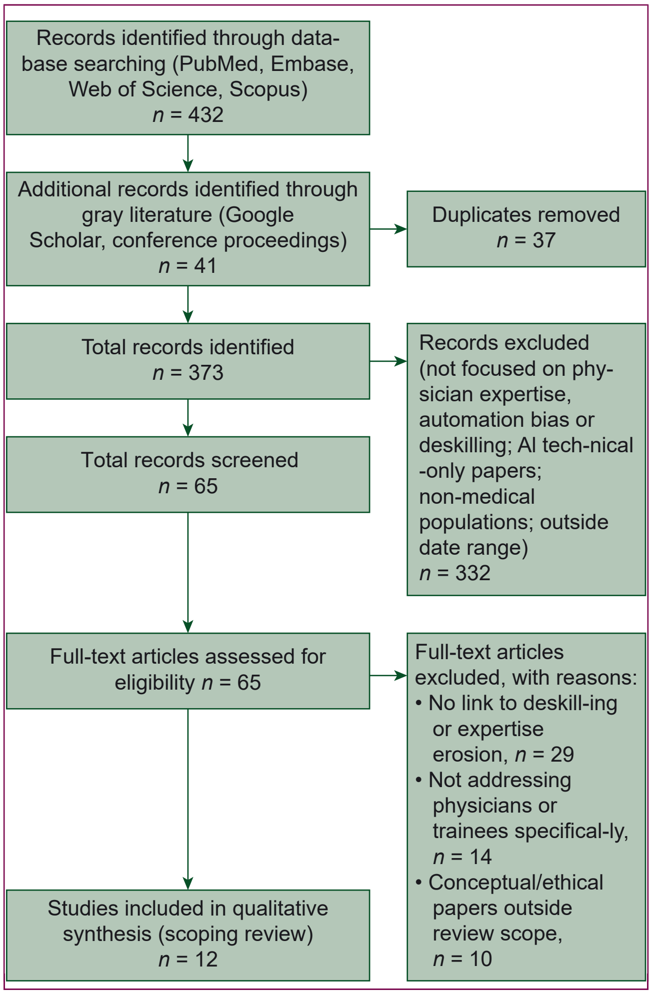

Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flowchart.

Gastroenterology & Hepatology in 2025 analyzed >23 000 procedures.[15] Endoscopists' adenoma detection rate (ADR) decreased from 28.4% before AI introduction to 22.4% when working without AI after routine AI exposure, while ADR remained at 25.3% with AI support. This provided direct evidence of behavioral dependence and skill erosion after repeated AI use.

Radiology

In breast imaging, a controlled reader study published in 2023 evaluated 27 radiologists interpreting 720 mammograms with and without AI support. When AI provided incorrect recommendations, radiologists' error rates increased by 12%-15%, even among experienced readers, demonstrating a clear manifestation of automation bias.[16] A recent empirical study conducted during the coronavirus disease 2019 (COVID-19) pandemic evaluated radiology residents interpreting chest X-rays with varying degrees of AI support.[12] This study evaluated 32 radiology residents trained with an AI-assisted module for detecting COVID-19 pneumonia on chest X-rays. While AI-supported training resulted in a 12% improvement in sensitivity (from 0.73 to 0.85, $P < 0.01$) and an 18% reduction in mean reading time per case (from 68 s to 56 s), the authors highlighted the unresolved concern of whether long-term reliance on AI could erode independent diagnostic reasoning. Although performance remained stable at 3-month follow-up, indicating a short-term upskilling effect with specificity maintained at ~0.80, the study underscores the importance of longitudinal evaluation to rule out potential deskilling over time.

Pathology

Evidence from computational pathology further supports the risk of automation-induced deskilling. An experimental study involving 28 pathologists found that under time pressure, participants abandoned 7% of initially correct judgments when faced with erroneous AI suggestions, and the severity of diagnostic errors increased.[17] Beyond computational pathology, structural changes in laboratory medicine also illustrate deskilling mechanisms: following the transition to human papillomavirus primary screening in the UK, cytology workloads declined by 80%-85%, raising concerns about the erosion of morphological expertise and the reduction of training opportunities for junior pathologists.[18,19]

Clinical decision support systems

The integration of AI into clinical decision support systems (CDSS) has raised significant concerns about automation bias and the erosion of independent reasoning. In a 2023 JAMA editorial, Khera et al. highlighted the risks of overreliance on automated recommendations, alert fatigue, and diminished clinical vigilance, warning that such dynamics may compromise physicians' ability to critically appraise AI outputs.[20] Abbasi and Hswen further emphasized that automation bias can systematically distort judgment, calling for robust evaluation frameworks to preserve clinical autonomy.[21] Empirical evidence confirms these risks. In a randomized crossover study of 40 clinicians diagnosing anterior cruciate ligament (ACL) rupture on magnetic resonance imaging, Wang et al. reported that AI assistance improved overall diagnostic accuracy from 87.2% $\pm$ 13.1% to 96.4% $\pm$ 1.9% ($P < 0.001$). However, 45.5% of errors under AI assistance were attributable to automation bias, affecting clinicians across all levels of experience. The authors proposed an 'AI suppression' strategy, which selectively withheld outputs with high misleading probability, reducing automation bias by 41.7%.[22] A complementary systems-level analysis by Patel et al. used a Bowtie risk framework to map the causes and consequences of automation bias in CDSS.[23] Their findings showed that interface design, workflow integration, and organizational policies are major contributors to deskilling risks, underscoring that the problem extends beyond individual cognition to organizational and interface-related contributors.

Cross-specialty reviews and professional competencies

A 2025 systematic review synthesized 58 studies examining the competencies physicians require to use AI in clinical

Table 1. Key studies on deskilling in artificial intelligence

Study	PubMed URL	Specialty	Main findings	Deskilling risk
Wang et al. (2023)[23] (JAMIA, PMID: 37561535)	https://pubmed.ncbi.nlm.nih.gov/37561535/	Radiology (ACL tears on MRI)	AI improved accuracy from 87% to 96%, but 45% of errors due to automation bias	Loss of independent judgment
Savardi et al. (2025)[12] (Insights Imaging, PMID: 39881013)	https://pubmed.ncbi.nlm.nih.gov/39881013/	Radiology (COVID-19 CXR)	12% sensitivity gain, 18% faster readings, performance stable at 3 months	Cognitive offloading, risk of shallow reasoning
Tschandl et al. (2020)[33] (Lancet Digital Health, PMID: 32572267)	https://pubmed.ncbi.nlm.nih.gov/32572267/	Dermatology	AI support improved diagnostic accuracy but induced automation bias	Overreliance on AI suggestions
Ainechi et al. (2022)[34] (Endoscopy, PMID: 35605150)	https://pubmed.ncbi.nlm.nih.gov/35605150/	Gastroenterology (polyp detection)	AI increased adenoma detection rates, reduced variability across endoscopists	Reduced training in visual recognition skills
Bellahsen-Harrar et al. (2025)[35] (Mod Pathol, PMID: 40875775)	https://pubmed.ncbi.nlm.nih.gov/40875775/	Pathology (computational)	28 pathologists, 7% of correct initial diagnoses reversed due to misleading AI	Erosion of independent judgment
Rebolj et al. (2022)[36] (Cytopathology, PMID: 35377967)	https://pubmed.ncbi.nlm.nih.gov/35377967/	Pathology (HPV screening shift, UK)	National cytology practice: workload ↓ 80%-85%, laboratories reduced from 45 to 8	Reduced training opportunities

ACL, anterior cruciate ligament; COVID-19, coronavirus disease 2019; CXR, chest X-ray; HPV, human papillomavirus; MRI, magnetic resonance imaging.

practice. It found broad consensus that AI will transform medical work, demanding both digital skills and human-centered competencies. However, descriptions of these competencies were often vague, inconsistent, and lacking concrete guidance. Major gaps include who should lead training, how it can fit within clinical workflows, and whether AI truly delivers promised benefits such as freeing physicians' time. The authors call for clearer frameworks and research to balance technical proficiency with essential human skills.[24] A qualitative study presented at the 2023 Australian Bioethics Conference analyzed interviews with clinicians, technologists, regulators, and developers on AI in diagnosis and screening.[25] Stakeholders expressed divergent views: some saw AI as causing harmful deskilling through overreliance and loss of core skills, while others framed it as necessary upskilling that frees clinicians for more complex tasks.

Other specialties

In surgical domains, over the past 25 years, robotic surgery has expanded into almost all specialties, with 2.6 million procedures carried out in 2024, doubling since 2019. This expansion has created a strong demand for training, as robotic surgery requires specific cognitive, psychomotor, and team-based skills not easily transferable from traditional approaches.[26] While several assessment tools such as Global Evaluative Assessment of Robotic Skills, Robotic Objective Structured Assessment of Technical Skills, and Robotic operative score have been developed, there is still no standardized curriculum, leading to significant variability in training quality. New technologies including AI, machine learning, and immersive simulations promise to personalize and enhance training. However, the growing reliance on automation and AI-based guidance raises concerns about deskilling, as fewer opportunities for manual practice may erode surgeons' tactile dexterity and independent decision making if not counterbalanced by robust hybrid training models (Table 1).

Cross-cutting patterns and plausible drivers

Across the empirical studies, several recurring patterns and plausible drivers emerge that help explain when AI support may drift from short-term upskilling to dependence and performance erosion:

- Automation bias is most pronounced when cases are ambiguous/low prevalence, under time pressure, or when interfaces encourage default acceptance without requiring justification or independent verification.
- Repeated exposure to AI can reduce independent verification behaviors and weaken feedback loops (fewer deliberate checks, less reflective reasoning), leading to measurable performance drops when AI is withdrawn or unavailable.
- Training ecology effects: AI triage and workflow redesign can reduce exposure to foundational or 'easy' cases that are essential for skill consolidation; structural volume reductions (e.g. screening pathway changes) can also erode opportunities for practice and supervision.
- Organizational factors (workflow integration, incentives, lack of calibrated uncertainty/override feedback) can amplify dependence, even when the underlying model is accurate on average.

These patterns informed the mitigation levers discussed below (training design, interface choices, monitoring of competence drift, and governance).

DISCUSSION

Our synthesis shows that while AI can improve short-term performance, clear signals of deskilling are emerging. Wang et al.[22] demonstrated that, despite increasing diagnostic accuracy in ACL tears from 87% to 96%, nearly half of the errors were due to automation bias, reflecting a decline in independent judgment. Survey studies further indicate reduced confidence when AI outputs are available, suggesting progressive loss of self-reliance. Even in training contexts, such as Savardi et al.,[12] where residents improved sensitivity by 12% and reduced reading time by 18%, gains may conceal cognitive offloading, with faster decisions replacing deeper reasoning. Similar concerns have been raised in gastroenterology, where AI-assisted polyp detection risks diminishing trainees' ability to develop visual recognition skills. Overall, these findings confirm that deskilling is not theoretical but already detectable, raising concerns about the long-term preservation of core clinical competencies.

Oncology is likely to be particularly exposed to deskilling mechanisms because decision making is increasingly mediated by algorithmic tools across the full care pathway: imaging triage and response assessment, digital pathology, radiotherapy planning, and clinical decision support for systemic therapies. While these systems can improve standardization and safety, they may also shift clinicians from primary interpretation and planning toward supervisory monitoring, increasing the risk of automation bias when outputs are incorrect or applied outside the training distribution. In multidisciplinary tumor boards, overreliance on algorithmic recommendations may also narrow deliberation and reduce the visibility of tacit reasoning.

Philosophical contributions complement empirical findings by framing the deskilling debate more broadly. Palmer and Schwan argued that AI risks undermining the three classical pillars of medical expertise—techne (practical skills), episteme (scientific knowledge), and ethos (moral character)—and proposed a participatory, deliberative, and conservative approach to preserve them.[27] Similarly, Sparrow and Hatherley highlighted paradoxes of 'deep medicine', cautioning that while AI is often presented as a way to re-humanize care, institutional and economic pressures may in fact exacerbate dependence on technology and accelerate the erosion of clinical skills.[28] Taken together, these analyses reinforce concerns that the benefits of AI may come at the cost of a subtle but significant deskilling of physicians, underscoring the need for robust strategies to maintain both technical expertise and the moral dimension of medical practice.

In response to these emerging risks, strategies to counteract deskilling must be considered at both educational and institutional levels. Firstly, curricula should integrate AI literacy while reinforcing traditional clinical reasoning and hands-on training, ensuring that automation supplements rather than replaces core skills. Simulation-based learning and hybrid practice models, already explored in robotic surgery, could provide a blueprint for preserving manual and interpretative expertise in radiology, pathology, and gastroenterology.[29] Secondly, assessment frameworks must evolve to include not only accuracy and efficiency metrics, but also the maintenance of competencies such as diagnostic independence, procedural dexterity, and ethical decision making. Finally, institutional policies should balance the benefits of AI adoption with safeguards to maintain physicians' autonomy, emphasizing continuous professional development and encouraging critical engagement with AI outputs.[30] These measures are crucial to transform the integration of AI into a process of upskilling rather than a drift towards deskilling.

Our analysis has several limitations that warrant consideration. Firstly, most of the available evidence on deskilling remains fragmented across specialties, with heterogeneous study designs and variable methodological rigor. Many reports are based on expert opinion or small-scale qualitative investigations rather than large multicenter trials, which restricts the generalizability of findings. Secondly, the rapidly evolving nature of AI tools means that conclusions drawn today may not fully apply to the next generation of systems, where improved transparency, explainability, and adaptive training features could alter the balance between upskilling and deskilling. Thirdly, the diversity of clinical contexts—ranging from radiology to gastroenterology and surgery—makes it difficult to establish a unified framework for evaluating skill erosion across disciplines.

Future research should therefore aim at generating longitudinal and prospective data to better capture the trajectory of clinical competence in environments where AI is deeply integrated.[31] Cross-specialty comparative studies could clarify whether certain domains are more vulnerable to skill attrition. Additionally, developing validated assessment tools for measuring deskilling, alongside simulation-based interventions, will be crucial for designing targeted mitigation strategies.[22] Finally, embedding ethical and philosophical frameworks into empirical studies will enrich the understanding of which clinical skills are essential to preserve and which may be safely delegated to AI without compromising patient care.[32]

Conclusion

AI is reshaping medicine with unprecedented promise, yet clear signals of deskilling reveal that expertise cannot be taken for granted. Preserving the art and science of clinical judgment will require deliberate choices: embedding AI literacy into training, enforcing regulatory safeguards, and fostering genuine human—AI collaboration. Without such action, progress risks eroding the very expertise on which safe and compassionate patient care depends.

FUNDING

None declared.

DISCLOSURE

The authors have declared no conflicts of interest.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

This document was proofread using a generative AI tool (Copilot) for spelling and grammar corrections.

DATA SHARING

No datasets were generated or analyzed during the current study. This manuscript is based on a structured literature review and does not involve primary data collection. All sources referenced are publicly accessible and appropriately cited in the References section.

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